

Partial Derivatives and Notation

Suppose

$$f(x, y) = x + y$$

Then

$$\frac{\partial f}{\partial x} = 1$$

Change of Variables

Let

$$x = u, \quad y = u + v$$

Then

$$f = x + y = u + (u + v) = 2u + v$$

So,

$$\frac{\partial f}{\partial u} = 2$$

Thus, even though $x = u$, we see that

$$\frac{\partial f}{\partial x} \neq \frac{\partial f}{\partial u}$$

Important Note

We must specify what is held constant.

$$\left(\frac{\partial f}{\partial x}\right)_y \quad \text{means: keep } y \text{ constant}$$

$$\left(\frac{\partial f}{\partial u}\right)_v \quad \text{means: keep } v \text{ constant}$$

Hence,

$$\left(\frac{\partial f}{\partial x}\right)_y \neq \left(\frac{\partial f}{\partial x}\right)_v = \left(\frac{\partial f}{\partial u}\right)_v$$

Example: Area of a Triangle

The area of a triangle is

$$A = \frac{1}{2}ab \sin \theta$$

Assume it is a right triangle. Then

$$a = b \cos \theta$$

We study the rate of change of A with respect to θ .

1) Treat a, b, θ as independent

Keep a and b fixed:

$$\frac{\partial A}{\partial \theta} = \left(\frac{\partial A}{\partial \theta} \right)_{a,b} = \frac{1}{2} ab \cos \theta$$

2) Keep a constant

Then b depends on θ . From

$$a = b \cos \theta \quad \Rightarrow \quad b = \frac{a}{\cos \theta}$$

So,

$$A = \frac{1}{2} a \left(\frac{a}{\cos \theta} \right) \sin \theta$$

$$A = \frac{1}{2} a^2 \tan \theta$$

Thus,

$$\left(\frac{\partial A}{\partial \theta} \right)_a = \frac{1}{2} a^2 \sec^2 \theta$$

3) Keep b constant

Then a depends on θ :

$$a = b \cos \theta$$

So,

$$A = \frac{1}{2} (b \cos \theta) b \sin \theta = \frac{1}{2} b^2 \sin \theta \cos \theta$$

Thus,

$$\left(\frac{\partial A}{\partial \theta} \right)_b = \frac{1}{2} b^2 (\cos^2 \theta - \sin^2 \theta)$$

Compute $\left(\frac{\partial A}{\partial \theta} \right)_a$

Method 1: Substitution

From the constraint:

$$a = b \cos \theta \quad \Rightarrow \quad b = \frac{a}{\cos \theta} = a \sec \theta$$

Area:

$$A = \frac{1}{2} ab \sin \theta$$

Substitute b :

$$A = \frac{1}{2} a \left(\frac{a}{\cos \theta} \right) \sin \theta = \frac{1}{2} a^2 \frac{\sin \theta}{\cos \theta} = \frac{1}{2} a^2 \tan \theta$$

Differentiate:

$$\left(\frac{\partial A}{\partial \theta}\right)_a = \frac{1}{2}a^2 \sec^2 \theta$$

Method 2: Differentials

Keep a fixed:

$$da = 0$$

Constraint:

$$a = b \cos \theta$$

Differentiate:

$$da = \cos \theta db - b \sin \theta d\theta$$

Since $da = 0$:

$$0 = \cos \theta db - b \sin \theta d\theta$$

So:

$$\cos \theta db = b \sin \theta d\theta$$

$$db = b \tan \theta d\theta$$

Function

$$A = \frac{1}{2}ab \sin \theta$$

$$dA = \frac{1}{2}b \sin \theta da + \frac{1}{2}a \sin \theta db + \frac{1}{2}ab \cos \theta d\theta$$

So,

$$dA = \frac{1}{2}a \sin \theta (b \tan \theta d\theta)$$

$$= \frac{1}{2}ab \cos \theta d\theta$$

$$= \frac{1}{2}ab \frac{(\sin \theta \tan \theta + \cos \theta)}{\sec \theta} d\theta$$

$$\Rightarrow \left(\frac{\partial A}{\partial \theta}\right) = \frac{1}{2}ab \cos \theta$$

Summary

1. Write dA in terms of $da, db, d\theta$
2. If $a = \text{const} \Rightarrow da = 0$
3. Differentiate constraints $\Rightarrow$ solve for db in terms of $d\theta$
4. Plug into dA to get the answer

Chain Rule

$$\left(\frac{\partial A}{\partial \theta}\right)_a = A_\theta \left(\frac{\partial \theta}{\partial \theta}\right)_a + A_b \left(\frac{\partial b}{\partial \theta}\right)_a$$

Since $a = \text{const}$, we treat a as constant.

Lecture 15: Partial Differentials

Topics (Example 2)

- Function of several variables
- Partial derivatives:

$$f_x = \frac{\partial f}{\partial x}$$

- Gradient:

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

- Approximation:

$$\begin{aligned}\Delta f &\approx f_x \Delta x + f_y \Delta y + f_z \Delta z \\ &= \nabla f \cdot \Delta \mathbf{r}\end{aligned}$$

Tangent Plane

Tangent plane to surface:

$$f(x, y, z) = c$$

Normal vector:

$$\nabla f$$

Partial Differential Equations

(Equations involving partial derivatives of an unknown function)

Heat Equation

$$\frac{\partial f}{\partial t} = k \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right)$$

$f(x, y, z, t)$ = temperature at (x, y, z) at time t

Min/Max Problems

Critical points occur when:

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0$$

Second derivative test for functions of two variables.

Differentials

$$df = f_x dx + f_y dy + f_z dz$$

Chain Rule

If:

$$x = x(u, v), \quad y = y(u, v), \quad z = z(u, v)$$

Then:

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial u}$$

Non-Independent Variables and Constrained Derivatives

Non-independent variable:

$$g(x, y, z)$$

Min/Max problems: maximize or minimize a function f subject to a constraint:

$$g = c$$

Lagrange Multiplier Method

$$\nabla f = \lambda \nabla g$$

$$\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g = c \end{cases}$$

Constrained Partial Derivatives

Given:

$$f(x, y, z) \quad \text{where} \quad g(x, y, z) = c$$

Find:

$$\left(\frac{\partial f}{\partial z} \right)_y$$

Conditions:

$$y = \text{constant}, \quad z \text{ varies}, \quad x = x(y, z)$$

Rate of Change of f with Respect to z

Using differentials:

$$df = f_x dx + f_y dy + f_z dz$$

$$dg = g_x dx + g_y dy + g_z dz = 0$$

Derivation Using Differentials

From the constraint:

$$dg = g_x dx + g_y dy + g_z dz = 0$$

Holding y constant ($dy = 0$):

$$g_x dx + g_z dz = 0$$

So:

$$dx = -\frac{g_z}{g_x} dz$$

$$\left(\frac{\partial x}{\partial z} \right)_y = -\frac{g_z}{g_x}$$

Substitute into df :

$$df = f_x dx + f_z dz$$

$$df = f_x \left(-\frac{g_z}{g_x} dz \right) + f_z dz$$

$$df = \left(-\frac{f_x g_z}{g_x} + f_z \right) dz$$

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Using Chain Rule

$$\left(\frac{\partial f}{\partial z}\right)_y = \frac{\partial f}{\partial x} \left(\frac{\partial x}{\partial z}\right)_y + \frac{\partial f}{\partial z}$$

From constraint:

$$0 = \left(\frac{\partial g}{\partial z}\right)_y = g_x \left(\frac{\partial x}{\partial z}\right)_y + g_z$$

So:

$$\left(\frac{\partial x}{\partial z}\right)_y = -\frac{g_z}{g_x}$$

Final result:

$$\begin{aligned} \left(\frac{\partial f}{\partial z}\right)_y &= \frac{\partial f}{\partial x} \left(-\frac{g_z}{g_x}\right) + \frac{\partial f}{\partial z} \\ &= -\frac{f_x g_z}{g_x} + f_z \end{aligned}$$

Directional Derivatives

Also need to know directional derivatives.

Double Integrals

Recall: For a function of one variable,

$$\int_a^b f(x) dx = \text{area under the graph of } f \text{ over } [a, b]$$

Definition of Double Integral

Double integral represents volume under the surface:

$$z = f(x, y)$$

over a region R in the xy -plane.

$$\iint_R f(x, y) dA$$

Definition: Partition R into small pieces of area ΔA_i :

$$\sum_i f(x_i, y_i) \Delta A_i$$

Take the limit as $\Delta A \rightarrow 0$:

$$\iint_R f(x, y) dA$$

Computation Using Slices

To compute:

$$\iint_R f(x, y) \, dA$$

Take slices of the region. Let:

$$S(x) = \text{area of a slice at } x$$